Predicting Mental Wellness: A Regression Analysis

Comparative Study of OLS, Ridge, Lasso, Random Forest and XGBoost

Author: Mohamed Amine Laabidi

Course: Statistical Learning

Date: April 2026

Table of contents

Abstract....3
1. Introduction....3
2. Dataset Overview....3
2.1 Data Description:....3
2.2 Data Preprocessing & Cleaning:....4
Table 1: Variables Removed or Recoded During Preprocessing....4
3. Exploratory Data Analysis....5
3.1 Descriptive Statistics:....5
Table 2: Descriptive Statistics of Key Numeric Variables....5
3.2 Correlation Analysis:....6
Correlation matrix....6
3.3 Non-Linearity Assessment:....7
Residuals vs fitted plots....7
4. Methodology....8
4.1 Model Selection Rationale:....8
Table: Model selection....8
4.2 Train/Test Split & Cross-Validation Strategy:....8
4.3 OLS Linear Regression:....9
4.4 Ridge regression:....9
4.5 Lasso Regression:....9
4.6 Random Forest:....9
4.7 XGBoost:....9
5. Results....10
5.1 Model performance:....10
Table: Model Performance on Held-Out Test Set....10
Actual vs. Predicted....11
5.2 Best Model: Lasso (λ = 0.093):....11
6. Conclusions....12
6.1 Key Findings:....12

Abstract

Mental health is shaped by behavioral and demographic factors. In this report I identify the key predictors of a mental wellness index (0–100) using a survey of 400 participants, and compare five regression models on held-out data. Self-reported stress is by far the strongest predictor, followed by poor sleep quality, while work and leisure screen time show smaller negative associations. The linear models (OLS and Lasso) predict best, explaining about 91% of the variance on the test set; tree-based models do not improve on them.

1. Introduction

Mental wellness is increasingly recognized as a critical dimension of public health, with direct implications for productivity, quality of life, and healthcare burden. Despite growing awareness, the specific behavioral and lifestyle factors that most strongly predict individual wellness outcomes remain poorly quantified.

This project uses the public **Screen Time vs Mental Wellness Survey – 2025** dataset (Kaggle), with **400 survey participants** and **15 variables** capturing digital habits, sleep patterns, stress levels, and demographic information.

The objective is to:

- Identify the key behavioral and lifestyle predictors of mental wellness
- Evaluate and compare five regression models (OLS, Ridge, Lasso, Random Forest and XGBoost) on their predictive accuracy
- Select the best-performing model based on generalization to unseen data

2. Dataset Overview

2.1 Data Description:

The dataset contains:

- **400 observations**
- **15 variables**
- Target variable: **mental_wellness_index_0_100** a continuous score

Feature groups:

- **Demographics:** age, gender, occupation, work mode
- **Screen behavior:** total screen time, work screen hours, leisure screen hours
- **Sleep & stress:** sleep hours, sleep quality, stress level
- **Lifestyle:** exercise minutes per week, social hours per week, productivity score

2.2 Data Preprocessing & Cleaning:

I started by inspecting the dataset for missing values, type inconsistencies, and redundant variables. No missing values or duplicate rows were found. Besides the participant ID, three variables required action before modelling: two were removed and one was recoded.

Variable	Action	Reason
screen_time_hours	Removed	Linear combination of work_screen_hours + leisure_screen_hours
sleep_quality_1_5	Recoded to binary	Highly concentrated (68.8% scored 1/5; only 3 participants scored 4, none 5)
productivity_0_100	Removed	Near-leakage with target (r = 0.90) and strong multicollinearity with stress_level (r = −0.88)

Table 1: Variables Removed or Recoded During Preprocessing

Sleep Quality Recoding Decision:

The original sleep_quality_1_5 variable is highly concentrated: 68.8% of participants scored 1 out of 5, only 3 participants scored 4 and none scored 5. To avoid estimating effects from such sparse categories, the variable was collapsed into a **binary indicator** before modelling:

sleep_quality_binary = 1 → poor sleep quality (original score = 1)

sleep_quality_binary = 0 → some sleep quality (original score ≥ 2)

After cleaning, the dataset was split into training and test sets using an **80/20 ratio** with a fixed seed (42) for reproducibility, yielding 321 training observations and 79 test observations.

Preparation for modelling:

- Categorical variables (gender, occupation, work_mode) encoded using dummy coding

- Design matrices built with model.matrix() (intercept column removed) for XGBoost; caret applies the same dummy encoding internally for the other models

3. Exploratory Data Analysis

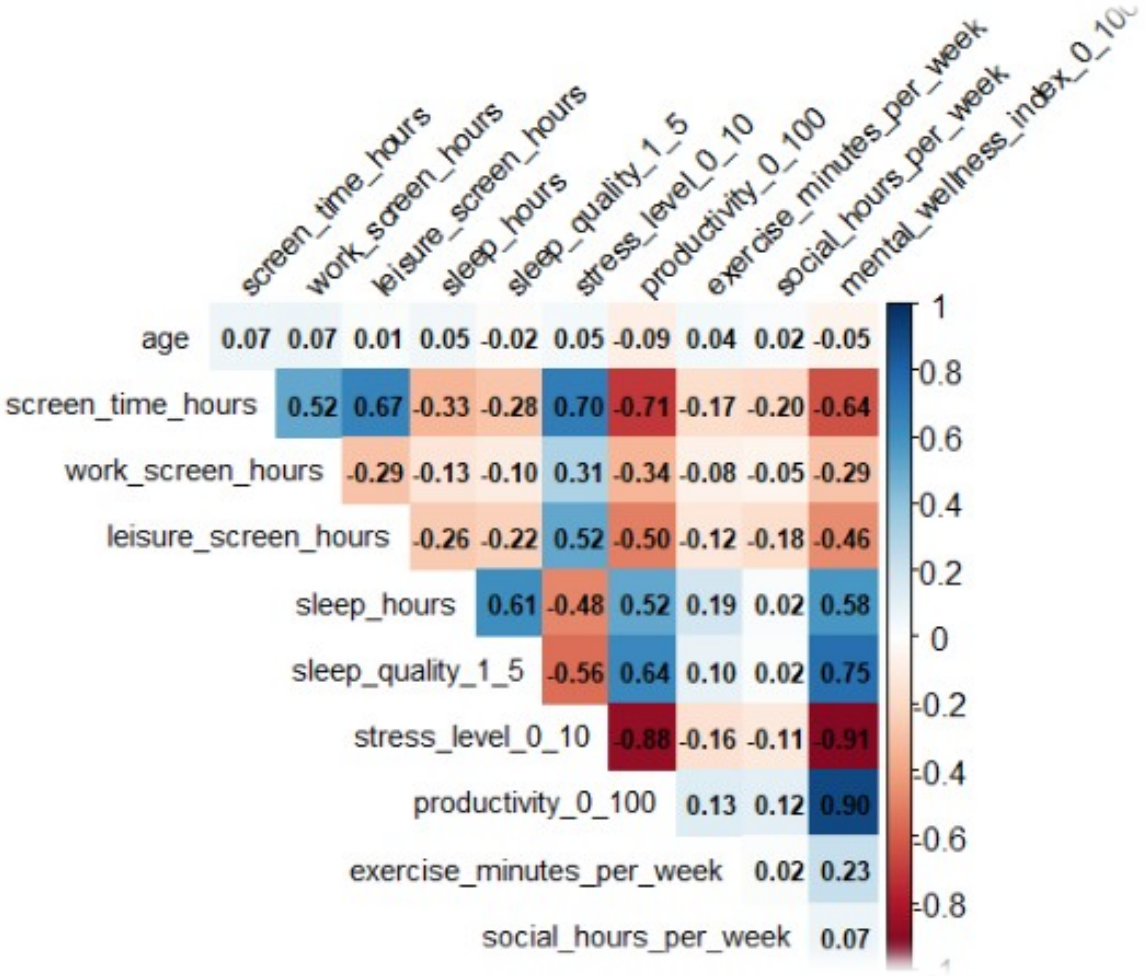
3.1 Descriptive Statistics:

Variable	Min	Q1	Median	Mean	Q3	Max
***a*ge**	16.00	24.00	30.00	29.78	35.00	60.00
work_screen_hours	0.11	0.70	1.46	2.18	3.01	12.04
leisure_screen_hours	0.89	5.46	6.70	6.84	8.42	13.35
sleep_hours	4.64	6.40	7.03	7.01	7.64	9.74
stress_level_0_10	0.00	6.90	8.80	8.15	10.00	10.00
mental_wellness_index	0.00	3.68	14.80	20.33	30.65	97.00

Table 2: Descriptive Statistics of Key Numeric Variables

Age ranges from 16 to 60 years with a median of 30, suggesting a sample skewed toward young working-age adults. Stress levels are notably high, with a median of 8.8 out of 10; a third of participants (33.8%) report the maximum value of 10. Sleep hours cluster tightly around 7 hours (mean = 7.01). Leisure screen time dominates over work screen time, with a mean of 6.84 hours per day compared to 2.18 hours for work-related usage. The target is strongly right-skewed (mean 20.3, median 14.8, skewness 1.22) and 15% of participants score exactly 0, a floor effect that a linear model cannot fully capture.

3.2 Correlation Analysis:



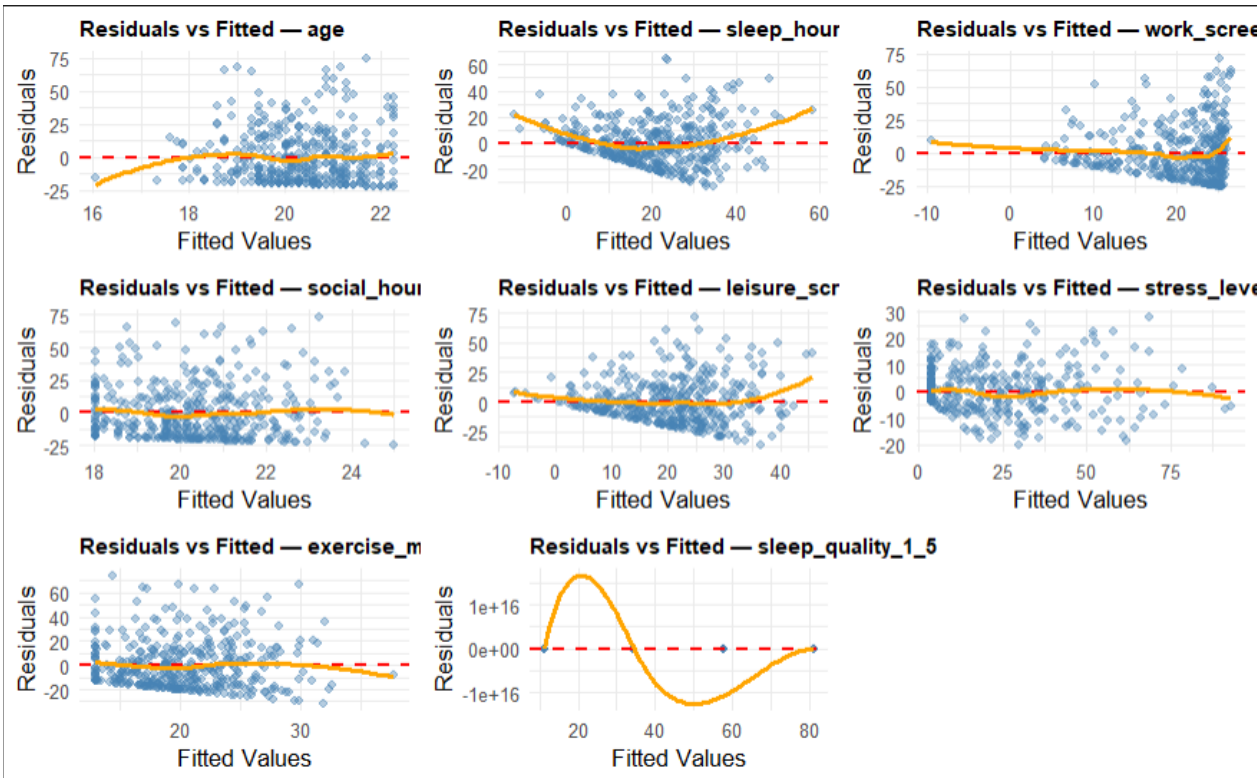
Correlation matrix

The correlation matrix reveals the following relationships with mental_wellness_index_0_100:

- ***stress_level_0_10**: Strong negative correlation **(r = −0.91)** the dominant predictor by a large margin*
- ***sleep_hours**: Moderate positive correlation **(r = 0.58)***
- ***leisure_screen_hours**: Moderate negative correlation **(r = −0.46)***
- ***work_screen_hours**: Weak negative correlation **(r = −0.29)***
- ***exercise_minutes_per_week**: Weak positive correlation **(r = 0.23)***
- ***age** and **social_hours_per_week**: Near-zero correlations **(r = −0.05 and r = 0.07)***

3.3 Non-Linearity Assessment:

Residuals vs fitted plots were generated for each numeric predictor. Non-linear patterns were detected in two variables:



Residuals vs fitted plots

- ***sleep_hours**: A visible U-shaped residual pattern, suggesting both too little and too much sleep are associated with lower wellness*
- ***leisure_screen_hours**: Mild but consistent bend in the loess curve, formally confirmed by a Ramsey RESET test (p = 0.032)*

These findings motivated the inclusion of tree-based models (Random Forest and XGBoost), which capture non-linear relationships without manual feature engineering. The performance gap between linear and tree-based models indicates how much non-linearity matters in this dataset.

4. Methodology

4.1 Model Selection Rationale:

Model	Role	Justification
OLS	Interpretable baseline	No regularisation exposes raw relationships between predictors and target
Ridge	Multicollinearity handler	L2 regularisation stabilises coefficient estimates when predictors are correlated
Lasso	Variable selector	L1 regularisation zeros out weak predictors
Random Forest	Non-linearity handler	Ensemble of decision trees, naturally immune to multicollinearity
XGBoost	Non-linearity handler	Gradient boosting captures non-linearity and feature interactions simultaneously

Table: Model selection

4.2 Train/Test Split & Cross-Validation Strategy:

The dataset was split 80/20 (321 train, 79 test) using a fixed random seed (42). OLS, Ridge, Lasso and Random Forest were evaluated with **10-fold repeated cross-validation (3 repetitions)** via the caret package; XGBoost's number of boosting rounds was tuned separately with xgb.cv() (10-fold). The CV RMSE estimates expected performance on unseen data. The difference Delta =

Test RMSE − CV RMSE is used as a rough overfitting check: a positive delta means the model did worse on the test set than cross-validation predicted. With only 79 test observations, small deltas largely reflect sampling noise.

4.3 OLS Linear Regression:

OLS serves as the fully interpretable baseline, estimating coefficients by minimizing the residual sum of squares. It is sensitive to multicollinearity and assumes strict linearity between predictors and target.

4.4 Ridge regression:

Ridge applies an L2 penalty $\lambda \sum_{j=1}^{p} \beta_j^2$ to the loss function, shrinking all coefficients toward zero without eliminating any. The optimal λ was selected via cross-validation over a log-spaced grid from 0.0001 to 100.

4.5 Lasso Regression:

Lasso applies an L1 penalty $\lambda \sum_{j=1}^{p} |\beta_j|$ which can shrink coefficients to exactly zero, performing automatic variable selection. The optimal λ was selected via the same cross-validation grid. The best λ found was 0.093.

4.6 Random Forest:

Random Forest builds an ensemble of 500 decision trees, each trained on a bootstrap sample with a random subset of predictors at each split. It naturally handles non-linearity and feature interactions. A first search with tuneRF() (out-of-bag error) suggested mtry = 6; the final grid search then used mtry $\in \{2, 3, 6, 7\}$ and min node size $\in \{1, 3, 5, 10\}$. The selected configuration was mtry = 7 with min node size = 5.

4.7 XGBoost:

XGBoost builds trees sequentially, with each tree correcting the residual errors of the previous one. The optimal number of boosting rounds (nrounds = 76) was determined via 10-fold cross-validation using xgb.cv(). Fixed hyperparameters: learning rate $\eta = 0.1$, max depth = 6, subsample = 0.8, colsample_bytree = 0.8.

5. Results

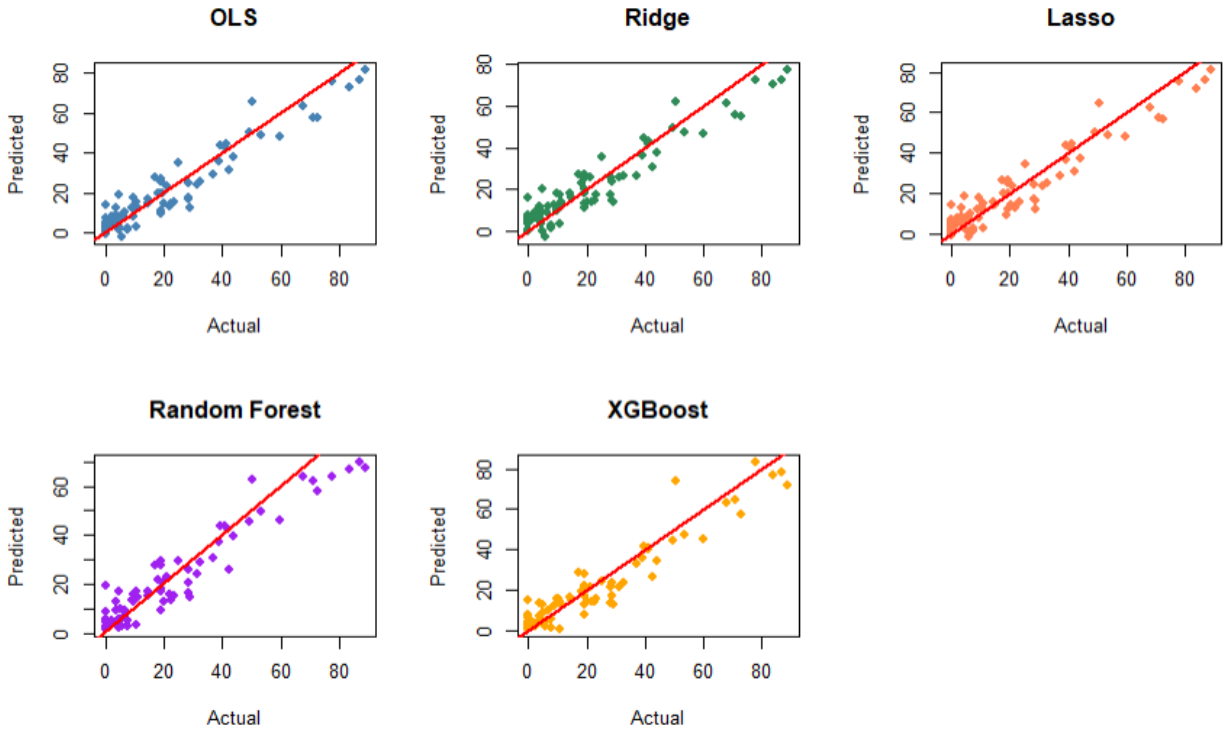
5.1 Model performance:

Model	R^2	RMSE	MAE	CV RMSE	Delta
OLS	0.910	6.999	5.732	6.322	+0.677
Ridge	0.895	7.532	6.224	6.461	+1.071
Lasso	**0.910**	**7.001**	**5.697**	**6.274**	**+0.728**
Random Forest	0.890	7.716	5.998	6.997	+0.719
XGBoost	0.900	7.357	5.541	7.916	−0.559

Table: Model Performance on Held-Out Test Set

All models explain between 89% and 91% of the variance in mental wellness on the test set, and none shows a large gap between CV and test error. The positive deltas of the first four models (+0.68 to +1.07 RMSE points) are moderate and, given the small test set, partly due to chance; Ridge, whose penalty shrinks every coefficient, has the largest one. XGBoost has a negative delta (−0.559), meaning it did slightly better on the test set than its cross-validation estimate; this is most likely sampling variability rather than a property of the model.

OLS and Lasso are virtually identical (RMSE 6.999 vs 7.001, $R^2 = 0.910$ for both). **Lasso is retained as the final model** because its penalty stabilises the coefficients under the moderate correlation between predictors, at no cost in accuracy; at the selected λ it removes only two dummy levels, so in practice it stays very close to OLS. That the tree-based models do not beat the linear ones indicates that the signal is **mostly linear**: the mild non-linearities found in Section 3.3 are not strong enough for the extra flexibility to pay off with 321 training observations.



Actual vs. Predicted

The red diagonal is the **perfect prediction line**: tighter clustering around it means better accuracy.

Random Forest is the weakest visually: its predictions rarely exceed 70 even when actual values reach 85, because averaging many trees pulls extreme predictions toward the mean.

OLS and Lasso are the strongest performers visually, with points tracking the diagonal closely across the full range (0–85). Their near-identical scatter is consistent with their matching R^2 of 0.910.

5.2 Best Model: Lasso (λ = 0.093):

Because predictors were standardised before fitting, each Lasso coefficient is the change in wellness associated with a **one-standard-deviation** increase in that predictor, holding the others constant. The dominant predictor is **stress_level_0_10** (−13.94 per SD; in original units the OLS fit gives about −6.8 wellness points per additional point of stress). **sleep_quality_binary** is the second strongest (−5.39 per SD; in the OLS fit, participants rating their sleep 1/5 score about 11.8 points lower than the others). **exercise_minutes_per_week** (+1.33) and **sleep_hours** (+0.86) are positively associated with wellness, while **work_screen_hours** (−1.16), **age** (−0.82) and **leisure_screen_hours** (−0.53) are negatively associated. **social_hours_per_week** (−0.14) and the demographic dummies are close to zero; Lasso sets genderMale and work_modeRemote exactly to zero.

6. Conclusions

6.1 Key Findings:

- ***Stress level is the dominant predictor****: about −6.8 wellness points per additional point of self-reported stress (OLS, $p < 0.001$), or −13.9 per standard deviation in the Lasso model.*
- ***Sleep quality matters beyond sleep duration****: participants rating their sleep 1/5 score about 11.8 points lower than others with the same sleep hours, stress level and screen time (OLS, $p < 0.001$).*
- ***Lasso is the selected final model*** *(RMSE = 7.001, $R^2 = 0.910$, MAE = 5.697): tied with OLS on accuracy and retained for the stability of its regularised coefficients.*
- ***Tree-based models do not improve on linear models****: Random Forest ($R^2 = 0.890$) and XGBoost ($R^2 = 0.900$) are slightly less accurate, suggesting the relationship is mostly linear and extra model complexity does not add value here.*
- ***Both screen time variables are negatively associated with wellness****: work_screen_hours (−1.16 per SD) and leisure_screen_hours (−0.53 per SD) are both retained by Lasso, but their effects are much smaller than those of stress and sleep.*
- ***Limitations****: the data are cross-sectional and self-reported, so all results are associations rather than causal effects; the wellness index has a floor at 0 (15% of participants), and results rest on a single 80/20 split with a 79-observation test set.*